

ORGANIC CHEMISTRY – CONDENSATION POLYMERISATION

KEY VOCABULARY

Condensation polymerisation: monomers join and a small molecule is lost.

Small molecule: usually water, lost each time monomers join.

Two functional groups: each monomer reacts at both ends.

Polyester: a condensation polymer (e.g. from a diol + diacid).

COMMON MISCONCEPTIONS

~~Condensation has only one product.~~

✓ It makes the polymer AND a small molecule (water).

~~It uses one type of monomer with a C=C.~~

✓ It uses monomers with two functional groups each.

~~Addition and condensation are the same.~~

✓ Condensation loses a small molecule; addition does not.

ADDITION VS CONDENSATION

Addition: 1 monomer (C=C), 1 product.

Condensation: 2 monomers (2 groups each).

A small molecule (water) is lost each join.

Example: diol + dicarboxylic acid → polyester + water.

Both build long-chain polymers.

1 DEFINE

State what is meant by **condensation polymerisation**. [1 mark]

2 TWO PRODUCTS

Name the small molecule usually lost in condensation polymerisation. [1 mark]

3 MONOMERS

Explain why condensation monomers need two functional groups. [2 marks]

4 COMPARE

Compare addition and condensation polymerisation. [3 marks]

Feature	Addition	Condensation
No. of monomer types		
No. of products		
Small molecule lost?		

5 FILL THE GAPS

Complete the sentence using the word bank. [2 marks]

Each monomer has ____ functional groups, and ____ is lost when they join.

two

water

one

carbon dioxide

6 EXAMPLE

Name one polymer made by condensation polymerisation. [1 mark]